

Programming Fundamental

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Lab 3 Assignment

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BSCS-1C

Exercise 1: Multi-line Output Using Escape Sequences

Objective

Demonstrate how to print multiple lines of text within a single printf() statement using the newline (\n) escape character in C.

Program Code

```c
#include <stdio.h>


int main() {
    // Print multiple lines using \n within a single string
    printf("Line 1: Hello, World!\nLine 2: Welcome to C programming.\nLine 3: Learning newlines.\n");
    return 0;
}
```

Explanation

- **#include <stdio.h>**: Includes the Standard Input Output library required for ***printf()***.
- **\n**: The newline control sequence. Whenever the compiler encounters *\n* inside a string literal, it instructs the output stream to advance the cursor to the beginning of the next line.

Expected Output

```
Line 1: Hello, World!
Line 2: Welcome to C programming.
Line 3: Learning newlines.
```

Exercise 2: Working with Format Specifiers

Objective

Demonstrate data formatting in C by outputting primitive data types (integers, floating-point numbers, and characters) using their respective printf() format specifiers.

```c
#include <stdio.h>

int main() {
    // Variable declarations and initializations
    int age = 25;
    float gpa = 3.85;
    char grade = 'A';

    // Output formatted values
    printf("Integer specifier (%%d): Age = %d\n", age);
    printf("Float specifier (%%f): GPA = %.2f\n", gpa);
    printf("Character specifier (%%c): Grade = %c\n", grade);

    return 0;
}
```

Format Specifiers Summary

Specifier	Data Type	Usage / Description
%d	int	Formats signed decimal integers.
%.2f	float	Formats floating-point numbers (rounded to 2 decimal places).

Specifier	Data Type	Usage / Description
%c	char	Formats a single character.
%%	N/A	Escapes the % character to print a literal percent sign.

Expected Output

```
Integer specifier (%d): Age = 25

Float specifier (%f): GPA = 3.85

Character specifier (%c): Grade = A
```

Exercise 3: Personal Introduction Program

Objective

Combine multi-line formatting and basic text printing to produce a structured personal summary output.

```c
#include <stdio.h>

int main() {
    printf("========================================\n");
    printf("          PERSONAL INTRODUCTION         \n");
    printf("========================================\n");
    printf("Name           : Minam\n");
    printf("Age            : 18\n");
    printf("Favorite Hobby: Researching New\n");
    printf("========================================\n");

    return 0;
}
```

Expected Output

```
========================================

        PERSONAL INTRODUCTION

========================================

Name         : Minam

Age          : 18

Favorite Hobby: Researching New

========================================
```